

Paper 1 — Expansion

Reframing to a Causal Design Using Staggered State Financial-Education Mandates

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Status: Proposal for decision — builds on completed & validated Paper 1 data cleaning

1. Executive Summary

Paper 1's data cleaning is complete and has passed independent validation. However, the five assembled datasets can support only **descriptive, correlational** analysis — they cannot substantiate the paper's central **causal** claim that financial-literacy education changes young people's financial behavior. This briefing proposes an expansion that reframes Paper 1 around a credible causal design: a **natural experiment** exploiting the fact that U.S. states adopted high-school financial-education graduation requirements at different times. Using the National Longitudinal Survey of Youth 1997 (NLSY97) linked to the history of these state "mandates," we can compare cohorts that were required to receive financial education against otherwise-similar cohorts that were not, and estimate the causal effect on adult saving, credit, and debt behavior. The literature review already supports this direction. Below: the rationale, the design, a six-package work plan with effort and ownership, the split between automatable and human-gated tasks, key risks, resource needs, a realistic timeline, and a low-cost feasibility check recommended **before** committing resources.

Headline: the active analyst effort is bounded ($\approx$ 4–6 weeks of work), but the overall calendar ($\approx$ 1–3 months) is driven almost entirely by one external item — approval for restricted geographic data — which is administrative lead time, not analytical effort.

2. Why the Current Framing Cannot Answer the Question

The research question is causal: does financial-literacy education *change* spending, saving, and credit behavior. Three features of the assembled data block that claim:

- **No treatment variable.** None of the five datasets (CFPB, SCF, NFCS 2018/2021, NLSY97) record whether a respondent actually received financial education. Without knowing who was "treated," we can only show that knowledge *correlates* with behavior — not that education *caused* it.
- **Population mismatch.** The question targets teens 13–22, but none of the datasets contain anyone under 18; they are adult surveys. The analysis sample is 18–24 *young adults*, not teens.
- **Non-comparable measures.** The same-named composite scores are built from different survey items in each dataset, so they are not a single construct that can be pooled or compared across sources.

Net effect: the cleaned data is well suited to *describing* how financial knowledge relates to money behavior among young adults, but it cannot isolate the *causal* effect of financial education, which is the paper's stated aim.

3. The Proposed Design: A Natural Experiment

The idea in plain terms. To prove education causes a change, we need two otherwise-similar groups — one that got financial education and one that did not. Over the years, some states passed laws requiring a personal-finance or economics course to graduate high school, and different states did so in different years. Whether a student received this education therefore depended largely on *which state they attended high school in and when* — circumstances outside the individual's control, approximating a natural “coin flip.”

How it identifies a causal effect. Because adoption was **staggered** across states and years, we can compare the adult financial behavior of cohorts that faced a requirement against comparable cohorts that did not, using a difference-in-differences / event-study framework (state and cohort fixed effects, standard errors clustered by state). The staggered timing is what lets us separate the effect of the mandate from general trends.

Outcomes. We measure adult financial behavior in later NLSY97 rounds — saving, debt, credit, and net worth. NLSY97 also fields a financial-literacy assessment at roughly ages 25–30, which becomes a legitimate knowledge outcome / mechanism (something the 18–24 framing had to drop).

Precedent. This is the established approach in the literature (Bernheim, Garrett & Maki 2001; Brown et al. 2016; Urban et al. 2020), which the Paper 1 lit review already supports — so the design is well grounded rather than novel-and-untested.

4. What Already Exists and Carries Over

This is an **extension**, not a restart. The following assets transfer directly:

- The reproducible cleaning pipeline and independent-validation harness.
- NLSY97 extract handling — codebook mapping, age anchoring, NLS missing-code logic, CPI deflation.
- The household income-to-poverty ($\leq 200\%$ FPL) underserved indicator.
- Pre-treatment control variables: demographics, education, and the **ASVAB** ability score (a strong control for prior aptitude, measured before the mandate exposure).

5. Work Plan

Six work packages. “Effort” is active analyst time; several packages run in parallel (see the critical path in Section 8).

#	Work package & objective	Est. effort	Owner	Depends on
WP 1	Restricted geographic data. Obtain NLSY97 high-school state & timing (not in the public file).	Low labor; weeks–months lead	PI / Institution	—
WP 2	Mandate database. Compile, by state and year, whether a personal-finance/econ course was required and the first affected graduating class.	1–2 weeks	Data team	—
WP 3	Treatment assignment. Merge WP1 & WP2 to label each respondent exposed / not exposed.	3–5 days	Data team	WP1, WP2
WP 4	Outcome extract & cleaning. Pull & clean later-round adult outcomes (saving, debt, credit, net worth) + the ~age 25–30 financial-literacy module.	~1 week	Data team	—
WP 5	Causal estimation. Difference-in-differences / event-study; state & cohort fixed effects; SEs clustered by state.	1–2 weeks	Lydia	WP3, WP4

#	Work package & objective	Est. effort	Owner	Depends on
WP 6	Robustness. Pre-trend/event-study plots, placebo tests, staggered-adoption estimators, few-cluster corrections.	~1 week	Lydia	WP5

Package detail

- **WP1** — The public NLSY97 hides state of residence to protect privacy. The linkage lives in the restricted Geocode file, released only under a signed data-security agreement with the survey agency (institutional sponsor required; possibly IRB). This is the project's gate.
- **WP2** — Sources are public and free (Council for Economic Education "Survey of the States," plus coding appendices in the precedent papers). The subtle part is dating: a law's passage year differs from the first graduating class it actually binds.
- **WP3** — Straightforward data work once WP1 & WP2 exist; handle students who moved states and those who did not finish high school.
- **WP4** — A new NLSY97 extract via the survey's download tool, then cleaned with the existing pipeline. Revives the financial-knowledge outcome via the older-age literacy module.
- **WP5 & WP6** — The estimation and the validity checks reviewers will expect; owned by Lydia with the data team supporting.

6. Division of Labor: Automatable vs. Human-Gated

Rule of thumb: the AI analyst can do anything that is "write code, crunch numbers, or compile public information" — fast. It **cannot** do things that require a legal signature, an interactive personal login, or human research-integrity judgment. Most of the *labor* is automatable; the *gates* are not.

Task	AI analyst?	Why
Compile public sources; draft the mandate database (WP2)	Yes (draft)	Web/code work — but must be human-verified against the actual laws
Sign the restricted-data application (WP1)	No	A legal agreement an accountable person/institution must sign
Download the NLSY97 extract (WP4 pull)	No	The survey's tool needs a personal login & interactive selection
Clean/merge/code the data (WP3, WP4-clean)	Yes	Pure scripting — fast once inputs exist
Write & run the estimation code (WP5)	Yes (execution)	But the specification choices are Lydia's to own
Verify mandate coding; own the identification decisions	No	Research-integrity judgment that must not be auto-rubber-stamped

7. Risks & Mitigations

Risk	Mitigation
Thin treatment variation (biggest research risk). Few states had mandates during the NLSY97 high-school window (~1998–2003), so too few “treated” respondents could leave the study underpowered.	Run the Section 9 feasibility check first . If variation is too thin, switch treatment source or pair NLSY97 with a complementary design (e.g., credit-panel or ACS-based) before committing.
Restricted-data lead time (WP1). Approval is external calendar time on the critical path.	File immediately; run WP2 and WP4 fully in parallel while waiting.
Mandate-coding validity . Wrong effective dates would bias the entire result.	Cross-check against published appendices; code passage vs. first-affected cohort explicitly; report sensitivity to alternative codings.
Staggered-adoption econometrics & few clusters . Naive two-way fixed effects can be biased; few treated states weaken clustered inference.	Use modern estimators (Callaway & Sant’Anna; Sun & Abraham); apply wild-cluster bootstrap for few clusters.
Migration & attrition . Students move; panels lose people.	Define exposure by high-school state; test robustness on movers; use survey weights and attrition checks.

8. Timeline & Critical Path

- **Active analyst effort:** ≈ 4–6 weeks total (much of it compressible by the AI analyst to days).
- **Calendar time:** ≈ 1–3 months — dominated by WP1 (restricted-data approval), which gates WP3 and WP5.
- **Parallelization:** WP2 (mandate database) and WP4 (outcome extract & cleaning) proceed immediately, independent of the WP1 approval queue, so no time is lost waiting.

Phase	Runs	Contents
Now (no gates)	Days	Feasibility check (Sec. 9); begin WP2; file WP1 application; begin WP4 pull
While WP1 pending	Weeks–months	Finish WP2 (verified) and WP4 (cleaned); pre-register the design
After WP1 approved	~2–3 weeks	WP3 merge → WP5 estimation → WP6 robustness
Write-up	Ongoing	Results into the Paper 1 draft alongside the lit review

9. Recommended Immediate Next Step: Feasibility Pre-Check

Before anyone files for restricted data, run a half-day, no-gates check: draft the mandate database (WP2) from public sources and estimate how many states had a financial-education requirement in effect during the NLSY97 high-school window (~1998–2003), and thus roughly what share of respondents would be “treated.” If that share is healthy, green-light WP1. If only a few states qualify, the study would be underpowered — and it is far better to learn that in half a day than after a multi-month data-access application. This single step de-risks the entire go/no-go decision.

10. Decision Requested from the Lead

- **(a)** Approve the half-day feasibility pre-check (Section 9)?
- **(b)** Identify an institutional sponsor able to file and sign the restricted-data (Geocode) application (WP1).

- **(c)** Confirm Lydia's availability for the estimation phase (WP5–WP6).
- **(d)** Confirm the outcome set and age window for the adult-outcome extract (WP4).

Appendix — Key References & Data Sources

- Bernheim, Garrett & Maki (2001), "Education and saving: The long-term effects of high school financial curriculum mandates," *Journal of Public Economics*.
- Brown, Grigsby, van der Klaauw, Wen & Zafar (2016), "Financial Education and the Debt Behavior of the Young," *Review of Financial Studies*.
- Urban, Schmeiser, Collins & Brown (2020), on the effects of state financial-education mandates.
- Council for Economic Education, *Survey of the States* (biennial) — mandate tracking source.
- Callaway & Sant'Anna (2021); Sun & Abraham (2021); Goodman-Bacon (2021) — staggered difference-in-differences methods.
- NLSY97 *Geocode* (restricted-access) — U.S. Bureau of Labor Statistics / NLS program.

Companion repository documents: docs/cleaning_log.md, docs/variable_dictionary_paper1.md, docs/analysis_decisions.md, docs/validation_report_paper1.md.